# Multimodal Skin Cancer Detection: Combining Image And Tabular Data With Deep Learning

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## 1. Problem Statement and Motivation

Skin cancer is the prominent type of cancer accounting for a third of all cancer diagnoses. Early detection is extremely important because patient outcome depends on it. When detected early survival rate is really high but when detection is late outcomes worsen significantly. Currently the main way to get diagnosed is through a dermatologist which many people do not have access to. If we can create a method for detecting skin cancer that is widely accessible then skin cancer can be detected earlier leading to more saved lives.

In this project we used ISIC 2024 Skin Cancer Detection Challenge data and problem set hosted on Kaggle. This challenge provides images along with tabular features such as patient demographics, lesion geometry, and color. We look at combining image information from the photographs with the tabular features to see if it improves classification performance. To achieve this we built multiple models and conducted ablation experiments to determine the value of each component.

## 2. Related Work

Deep learning has been used in skin lesion classification before. Convolutional neural networks trained on dermoscopic images could perform as well as dermatologist diagnosis shown by Esteva. Different models such as ResNet, DenseNet, and EfficientNet have been used as backbones for various medical imaging tasks.

EfficientNet is good for its predictive performance and computational power needed for it. Since medical datasets are rather small, transfer learning is used most of the time and usually results in better results than if you were to train from scratch.

Tabular data has been used by previous competitors in the ISIC 2024 Skin Cancer Challenge. They used boosted tree models in their approach.

We look to combine these previously done approaches together. We created an architecture where the model first learns the embeddings from images and tabular data separately and then combines them in order to create the final prediction.

## 3. Dataset Description

### 3.1 Data source

Our dataset comes from the ISIC 2024 Kaggle Challenge on skin cancer detection with 3D total photography. There are two main components to the data the images and the

metadata file. The metadata contains information such as patient information, lesion size, shape, color, etc.

### 3.2 Class Distribution

Like many other medical datasets there is a huge imbalance between positive (malignant) and negative (benign) samples.

| Class | Count |
|---|---|
| Benign | 400,666 |
| Malignant | 393 |

The dataset is extremely imbalanced with most of the data being benign. The malignant data only represents .1% of the data while benign represents 99.9% of the dataset. This imbalance makes it hard to train the model directly. We would need to balance the dataset somehow.

### 3.3 Tabular Features

We used 75 numeric features. Some of these features were from the raw metadata, others were data engineered. The raw features include age, lesion diameter, perimeter, area, etc. The engineered features capture relationships such as color differences between lesion and surrounding area, geometric ratios, normalized lesion size, and interaction terms

### 3.4 Handling Class Imbalance

In order to handle the class imbalance we used two methods. The first method was to downsample the overall amount of malignant data. We randomly choose for every one benign data point we include 20 malignant data points. We downsampled the amount of malignant data we had. The second method was to enforce balanced mini-batch training so each batch had equal amounts of benign and malignant data points. This reduces bias towards malignant data and made the classifier identify more benign cases.

### 3.5 Train, Validation, and Test Split

| Split | Approx. Samples | Purpose |
|---|---|---|
| Training | ~4,950 | Model fitting |

| | | |
|---|---|---|
| Validation | ~1,650 | Hyperparameter selection and early stopping |
| Test | ~1,650 | Final held-out evaluation |

We used Statifiedgroupkfold with grouping of patient identity in order to prevent leakage. This makes sure that a patient isn't seen in both the training and testing split.

## 4. Model Design

We implemented a multimodal network that combines image and tabular data. The image branch processes the dermoscopic images using EfficientNet-B0 pretrained on ImageNet. The original classification head was removed and the backbone outputs 1280-dimensional embedding. This was then combined with the tabular branch that was a two layer multilayer perceptron with batch normalization, ReLU activations, and dropout producing a 128-dimensional embedding from 75 cleaned features. We also evaluated several variants of this model. This includes the tabular only model, image only model, a multimodal version using average pooling instead of GeM pooling, and a version with EfficientNet backbone frozen during training. These ablations were crucial to understanding which contributed most to the final performance.

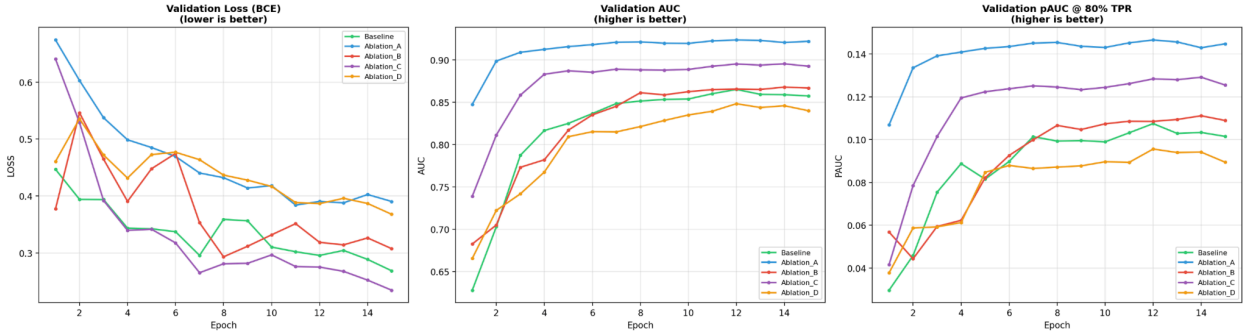
## 5. Training Procedure

| **Hyperparameter** | **Value** |
|---|---|
| Optimizer | Adam |
| Learning rate | 1 × 10^-4 |
| Weight decay | 1 × 10^-6 |
| Scheduler | Cosine annealing |
| Epochs | 15 |
| Train batch size | 32 |
| Validation/Test batch size | 64 |
| Loss function | Binary cross-entropy |

Training images were first resized to 224x224 and augmented using random flips, rotations, affine transforms, brightness adjustment, and color changes. These augmentations were only applied to the training set. Validation and test images were only normalized and resized.

All of the features were standardized with Z score normalization computed from the training data only. Missing values were replaced with zero after normalization.

## 6. Evaluation Metrics




We used AUC-ROC to evaluate our performance. AUC-ROC shows how well the model separates malignant from benign lesions across all classification thresholds. We also used Partial AUC which aligns with how the ISIC 2024 challenge was evaluating contestants. It emphasizes operating regions with high sensitivity.

## 7. Experimental Result

| Experiment | Test AUC | Test pAUC |
|---|---|---|
| Baseline multimodal | 0.9014 | 0.1264 |
| Tabular only | 0.9283 | 0.1469 |

| | | |
|---|---|---|
| Image only | 0.9058 | 0.1311 |
| Multimodal + AvgPool | 0.9391 | 0.1563 |
| Frozen backbone | 0.8817 | 0.1096 |

The results show that the multimodal model using average pooling performed the best in AUC and partial AUC on the test data. The tabular only model performed really well also and outperformed the image only model. This shows that the metadata contained strong predictive powers and is useful in determining between benign and malignant. Since the multimodal model performed the best we can see that image and tabular data is useful in determining skin cancer type.

## 8. Analysis and Discussion

One of the biggest findings from this study is that tabular features are extremely informative. This is consistent with findings from other competitors in the ISIC 2024 challenge. Structured metadata played a major role in high performing solutions. Many metadata features reflect meaningful characteristics in determining skin cancer type. Another notable result was that GeM pooling did not perform as well as average pooling. A theory is that GeM might have added flexibility that was not helpful under training regime and dataset size. Due to the nature of our data, average pooling was better. Fine-tuning the image backbone also proved important in our findings. The frozen backbone model had the lowest ROC. Dermoscopic images contain different characteristics than regular images found in ImageNet. Domain specific adaptation is necessary in order to produce good results.

## 9. Limitations and Future Work

This project has several limitations. First the dataset is extremely imbalanced so the amount of malignant examples remains small. Second we did not do a full cross-validation analysis so our estimations have variance. Third, our model is not super complex.

Future work includes obtaining more malignant data or upsampling the data by using some other methods. Another thing we could do is further explore larger backbones, ensemble methods, and other tabular models such as LightGBM. Lastly we could perform validation across multiple folds and do more systematic hyperparameter tuning.

## 10. Team Contributions

This project was completed by Ethan and Kelvin. Ethan did the data preprocessing, model design, training, and experimentation. Kelvin did the problem set up, analysis, and final report.

Github Link: https://github.com/Ethan-Shapiro/CS273P-final-project

## References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